BCCDC Public Health Lab *Mycobacterium tuberculosis* complex Antimicrobial Resistance Gene Detection Validation using TBProfiler

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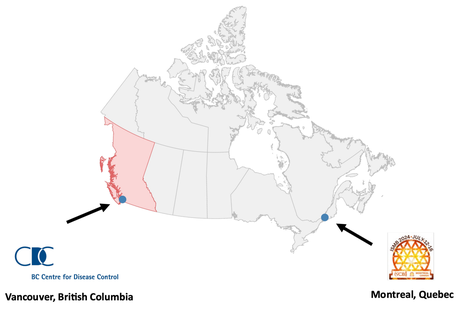
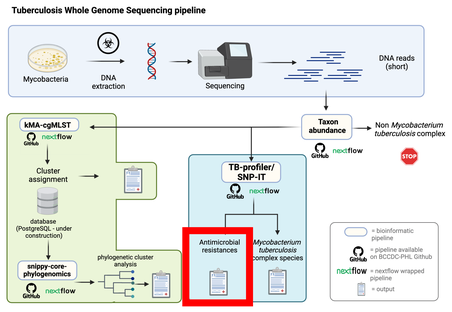
Background & Objective

- **In summer of 2023, BCCDC PHL TB/Mycobacteriology lab became the first provincial laboratory in Canada to transition to universal prospective whole genome sequencing (WGS)-based analysis for *M. tuberculosis* (MTB) genotyping. This approach allows concurrent genotyping, identification of drug resistance markers, and identification of MTB complex (MTBC) subspecies**
- **Phenotypic drug susceptibility testing (pDST) serves as the current gold standard for determining antimicrobial resistance (AMR) in MTBC. However, the prolonged turnaround time may delay treatment of tuberculosis patients.**
- **WGS is an alternative method, allowing for rapid detection of mutations associated with AMR. TBProfiler, developed by Jody Phelan, provides AMR predictions by comparing genetic variants of sequenced isolates to its drug-resistance database.**
- **To validate TBProfiler for use at the BCCDC Public Health lab, genotypic predictions were compared to pDST data for various antibiotics including Isoniazid, Rifampicin, and Fluoroquinolones. Isolates were classified based on concordance between phenotypic and genotypic results.**



BACTEC MGIT 960 Phenotypic Susceptibility Testing for MTB

Image: yumpu.com

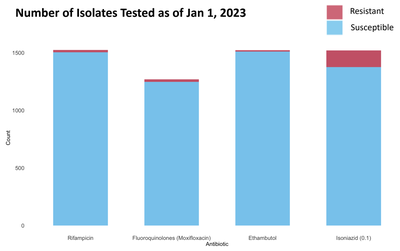


Methods

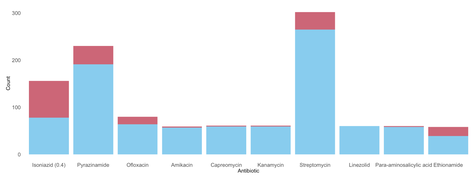
- **Phenotypic data was collected for 1776 isolates tested at BCCDC PHL TB Lab and the National Microbiology Laboratory for resistance to Isoniazid, Rifampicin, Ethambutol, Pyrazinamide, Fluoroquinolone(s), Amikacin, Capreomycin, Kanamycin, Streptomycin, Linezolid, Para-aminosalicylic acid and Ethionamide.**
- **Isolates resistant to Isoniazid at a critical concentration (CC) of 0.1 µg/mL were further tested at a CC of 0.4 µg/mL**

Clinical Phenotypic Susceptibility Data

Number of Isolates Tested as of Jan 1, 2023



Antibiotics with > 1000 isolate pDST data



Antibiotics with <300 isolate pDST data

TBProfiler Genotypic AMR Prediction Example

sample_id	drug	mutation
ERR046855	Rifampicin	rpoB p.His445Tyr
ERR046855	Isoniazid	katG p.Ser315Thr
ERR046855	Ethambutol	embB p.Met306Ile
ERR046855	Pyrazinamide	No mutation detected

- **TBProfiler (v.4.3.0_tbdb_4738132) was used to generate an AMR resistance prediction for the same isolates.**
- **The genotypic and phenotypic results were compared and assessed for accuracy, sensitivity, specificity and clinical validity**
- **The Positive Predictive Value (PPV) of the predicted mutations from the World Health Organization (WHO) Catalogue of mutations in MTB complex [4] was used to categorize the TBProfiler genotypic results by HIGH (>90%), MEDIUM (75-90%), LOW (<75%) and Not Available (NA) based on WHO PPV | SOLO (PPV conditional on mutation being solo i.e.. not in combination with other mutations)**

Results Classification		Predicted: TB-Profiler	
		No Mutation Detected -	Mutation Associated w/ Resistance +
Actual: pDST	Sensitive -	True Negative (TN)	False Positive (FP)
	Resistant +	False Negative (FN)	True Positive (TP)

WHO Catalogue of Mutations

Mutation named as described in the chapter: Detailed methods

Final confidence grading of a mutation

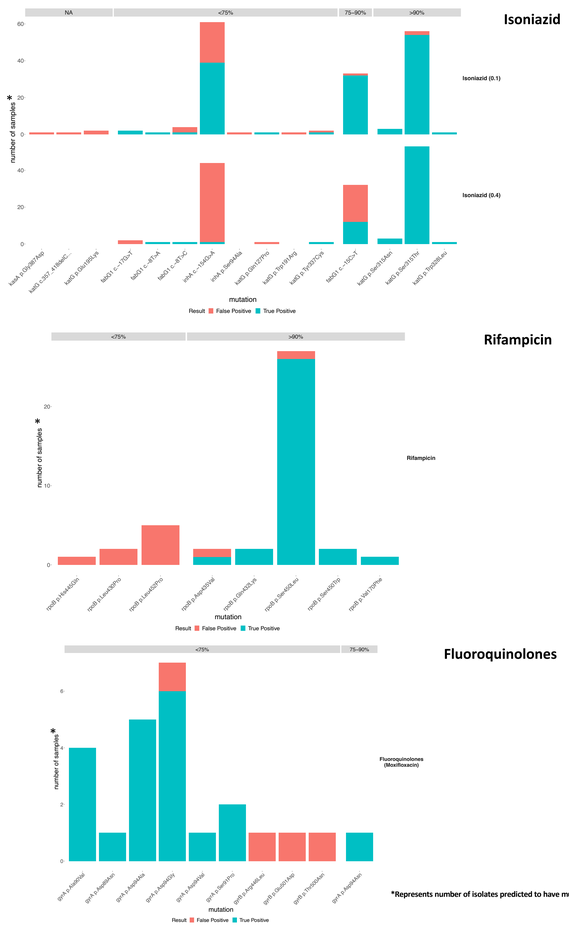
Drug	Variant (common name)	Present_S	Absent_S	Present_R	Absent_R	SENSITIVITY	SPECIFICITY	PPV	PPV | SOLO	PPV | SOLO, lb	PPV | SOLO, ub	OR SOLO	INITIAL CONFIDENCE GRADING	DATASET(S)	ADDITIONAL GRADING CRITERIA	FINAL CONFIDENCE GRADING
RIF	rpoB_S450L	74	24473	6536	3333	66.2%	99.7%	98.9%	98.6%	98.3%	98.9%	584.342	Assoc w R	ALL+WHO		1) Assoc w R
RIF	rpoB_L430P	103	24330	106	9743	1.1%	99.6%	50.7%	23.1%	16.3%	31.2%	0.806	Uncert. Sig.	ALL+WHO	Borderline	1) Assoc w R
RIF	rpoB_V695L	55	20207	52	6678	0.8%	99.7%	48.6%	1.8%	0.0%	9.6%	0.058	Not assoc w R	WHO		3) Not assoc w R

Drug in focus

Additional grading criteria applied when relevant to reach the Final confidence grading

Image: www.who.int/

Results



*Represents number of isolates predicted to have mutation conferring antibiotic resistance

- **The following results are highlighted for Rifampicin, Isoniazid and Fluoroquinolones (Moxifloxacin as current phenotypic option).**
- **The number of true positive (blue) and false positive (red) results based on the phenotypic data are compared for each resistance mutation predicted by TBProfiler based on WHO PPV range.**
- **Categorizing the results by high (>90%), medium (75-90%), and low (<75%) WHO PPV showed a clear trend of high PPV mutations having a greater number of true positive predictions for Rifampicin and Isoniazid.**
- **Discrepancies were observed between phenotypic and genotypic data, particularly for mutations associated with low-level isoniazid resistance.**
- **The same trend was not observed for Fluoroquinolones, where mutations with a low PPV were predominantly concordant with pDST data in our dataset.**

- **The WHO PPV | SOLO and validation status is appended to the mutations in the TBProfiler database so this data is output with each TBProfiler prediction to aid in result reporting**

sample_id	drug	mutation	WHO PPV|SOLO	Validation Status
123456	Isoniazid	fabG1 c.-8T>C	33.30%	SNP detected is known to confer low-level resistance
123456	Ethionamide	fabG1 c.-8T>C	20.00%	Mutation's definitive prediction of resistance has not been confirmed by BCCDC PHL's validation
123456	Isoniazid	katG p.Ser315Thr	98.60%	Predicted to be resistant

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Conclusions & References

- **The comparison between phenotypic and genotypic AMR data is highly dependent on the antibiotic, sample size, and specific mutations present within the validation dataset and database, which can be significantly influenced by the local TB epidemiology.**
- **The lack of mutation is insufficient to rule out antibiotic resistance. Phenotypic testing remains the gold standard for determination of antimicrobial resistance.**
- **Cases where a WHO-classified resistance-predicting mutation is detected in conjunction with a susceptible phenotypic result should undergo dedicated review by the microbiology laboratory and the clinical team.**
- **TBProfiler was validated at the BCCDC Public Health Lab as a predictive assay. Continued validation of new isolates' phenotypic drug susceptibility and mutations is required by clinical laboratories for optimized and reliable clinical application.**

Public Health Agency of Canada | Agence de la santé publique du Canada | BC Centre for Disease Control | Provincial Health Services Authority

1. https://www.yumpu.com/en/document/view/16478130/promotional-flyer-bd
2 https://github.com/jodyphelan/TBProfiler
3. https://github.com/BCCDC-PHL/tbprofiler-nf
4. WHO Catalogue of mutations: https://www.who.int/publications/i/item/9789240028173